

Kyra Riedel

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EDUCATION

Northeastern University, Boston, MA

Khoury College of Computer Sciences

Master's in Computer Science

August 2026

Graduate Coursework: Machine Learning, Computer Vision, Software Engineering, App Development, Cybersecurity

Bachelor of Science in Computer Science and Biology with a minor in Computational Data Analytics

May 2025

COMPUTER KNOWLEDGE

Languages: Java | Python | JavaScript | TypeScript | R | C++ | SQL

Frameworks/Tools: React | Express | Node.js | Firebase | PostgreSQL | MongoDB | Git | AWS

WORK EXPERIENCE

Millennium Management, New York, New York

June – August 2025

Equities Technologies Intern

- Built a mobile portfolio dashboard with 30+ interactive visualizations used by Portfolio Managers, handling pagination, rate limiting, and caching per-user and per-investment API calls, with lazy loading on scroll
- Rebuilt a legacy desktop codebase into a mobile-first version, stripping the desktop logic to essential pages/metrics and redesigning the UI (optimized for readability and interaction) for smaller screens

New York Genome Center, New York, New York

July 2023 – January 2024

Bioinformatics Analyst – Intern

- Optimized a PCA-based genomic denoising tool in R on 1000+ samples, then benchmarked it against 3 competing tools, identifying critical pipeline failures that informed a workflow redesign
- Built performance visualizations in ggplot2 comparing algorithm accuracy metrics and presented findings to 100+ researchers at the NY Genomics and Human Genetics Symposium (January 2024)

Northeastern University, Boston, Massachusetts

January 2022 – July 2024

Teaching Assistant for Algorithms and Data Structures

- Led weekly recitations for 20 students and one-on-one office hours for 100+ students

PROJECTS

MBTA Transit Analysis | Python, PostgreSQL

September 2026 – Present

- Built a PostgreSQL database and Python pipeline to load MBTA data (~849M rows) spanning 2020–2026, with varying schemas and a GTFS station lookup to map raw IDs to stop names

Image/Video Processor | C++

January – February 2026

- Developed a CLI pipeline that searches and ranks ink-layer combinations for a 6-ink risograph palette, exporting spot plates and print-ready files via CMake-wired OpenCV/ONNX Runtime dependencies
- Implemented subtractive layer compositing and a Photoshop-style GUI with face/depth detection

Patient Portal App | Java, Android, Firebase Auth, Realtime Database

October 2024 – January 2025

- Built a Firebase-backed backend for role-based patient and physician data, with real-time task and appointment sync and a repository layer for CRUD and indexed queries
- Implemented Firebase Auth (email/password, reset, session routing by role) and Cloud Storage for recovery photos linked to task records

Stack Overflow Adaptation | React, JavaScript, Express, MongoDB

February – May 2024

- Developed a question board REST API with MongoDB, modeling schemas for questions, answers, comments, tags, and accounts, while supporting search, sorting, voting, and author-only answer acceptance
- Computed reputation from post votes on profile read, and covered the API routes with Jest and Supertest